

Discrete Structures

Topic 2

Introduction to set theory

- Finite and Infinite Sets,
- Functions

Sets



§ 1

The Conception of Power or Cardinal Number

By an “aggregate” (*Menge*) we are to understand any collection into a whole (*Zusammenfassung zu einem Ganzen*) M of definite and separate objects m of our intuition or our thought. These objects are called the “elements” of M .

Figure: Georg Cantor (1845-1918); extract

What is a set?

- A **set** is an unordered collection of objects.
- The objects in a set are called its **elements**.

Formally:

Let P be a property. Any collection of objects that are defined by (or satisfy) P is a set, i.e., $S = \{x \mid P(x)\}$.

Examples and Properties

□ Let A, B be two sets:

- Equality $A = B$
- Subset $A \subseteq B$,
- Cartesian product $A \times B = \{(a, b) \mid a \in A, b \in B\}$
- Union $A \cup B = \{x \mid a \in A \text{ or } b \in B\}$
- Intersection $A \cap B = \{x \mid a \in A \text{ and } b \in B\}$
- Empty set ϕ ,
- Power set of $A = \mathbf{P}(A)$ = set of all subsets of A .
- If U is the universe, then the complement of A ,
 $\bar{A} = A^c = \{x \in U \mid x \notin A\}$.

□ What is the difference between $\{\emptyset\}$ and $\emptyset$?

- $\emptyset$ is the empty set (size 0).
- $\{\emptyset\}$ is a set containing the empty set as an element (size 1).

Paradox

A barber is a man in town who shaves all and only those who do not shave themselves.

Barber's paradox: Does the barber shave himself?

Let B be the barber. Does B shave B ?

If yes, then B shaves someone who shaves himself, contradicting "barber shaves only those who don't shave themselves."

If no, then B doesn't shave himself, so by the rule the barber must shave him — contradiction again.

Paradox

Russell's paradox:

Let **S** be the set of all sets that **do not contain themselves**
i.e., $S = \{X \mid X \notin X\}$

- If S belongs to itself, then according to the definition of S, it must be a set that **does not contain itself**.
- If S does not belong to itself, then it satisfies the condition for membership in S.

So we get:

$$S \in S \iff S \notin S$$

That's **Russell's Paradox**. If $S \in S$, then $S \notin S$ and if $S \notin S$, then $S \in S$! How to resolve this?

ZFC

Need for Axiomatic approach to set theory (ZFC)

Start with a few objects defined. Then for a set A and a property P , $S = \{x \in A \mid P(x)\}$ is a set.

Why does this definition get rid of Russell's paradox?

ZFC

If A is a set and $P(x)$ is a property, then $\{x \in A \mid P(x)\}$ is also a set.

Naive set theory: A single universal set U that contains all sets, including itself $\rightarrow$ leads to Russell's paradox.

ZFC: An infinite hierarchy of layers $(V_0, V_1, V_2, \dots, V_\alpha)$ — every set lives at some stage, but the total collection V is only a **proper class** (cannot be an element of anything), not a set.

Key difference: You can only carve out subsets from an already existing set i.e., “Take a set A , and inside it pick all x such that $x \notin x$ (set x is not an element of set x)”

But you cannot say: “Take all sets in the universe with property $x \notin x$ ” — because that would require a “set of all sets,” which ZFC forbids. ZFC gives **construction rules** that make new sets from existing ones.

Growth of Power Sets [ZFC steps have some more details]

In general, size of
the set grows as:
 $V_{n+1} = 2^{V_n}$ [2
power of set size in
previous layer]

Step 1: Start with the base set

$$X = \{1, 2\}$$

Step 2: Power set of X

$$P(X) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

This has 4 elements.

Step 3: Power set of $P(X)$

Now we want all subsets of $P(X)$.

Since $P(X)$ has 4 elements, its power set has $2^4 = 16$ elements.

So $P(P(X))$ is:

1. $\emptyset$
2. $\{\emptyset\}$
3. $\{\{1\}\}$
4. $\{\{2\}\}$
5. $\{\{1, 2\}\}$
6. $\{\emptyset, \{1\}\}$
7. $\{\emptyset, \{2\}\}$
8. $\{\emptyset, \{1, 2\}\}$
9. $\{\{1\}, \{2\}\}$
10. $\{\{1\}, \{1, 2\}\}$
11. $\{\{2\}, \{1, 2\}\}$
12. $\{\emptyset, \{1\}, \{2\}\}$
13. $\{\emptyset, \{1\}, \{1, 2\}\}$
14. $\{\emptyset, \{2\}, \{1, 2\}\}$
15. $\{\{1\}, \{2\}, \{1, 2\}\}$
16. $\{\emptyset, \{1\}, \{2\}, \{1, 2\}\} = P(X)$

How does all this matter ?

Each **layer** was built only from the stuff below it.

Layer 0: raw atoms (1,2,3)

Layer 1: subsets of A

Layer 2: sets made out of raw subsets

... and so on.

A set in **Layer 1** only contains numbers, so it can't contain *itself*.

A set in **Layer 2** might contain a set from Layer 1, but not itself (because at the moment of construction, "itself" isn't available yet).

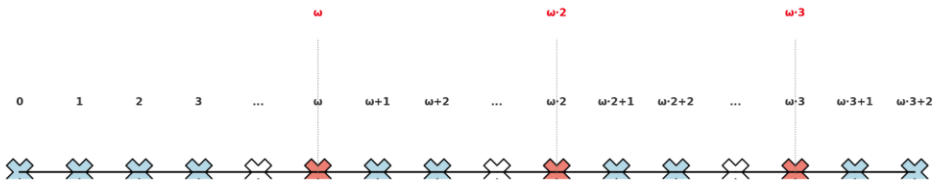
The same holds for Layer 3, 4, ... Every set is built strictly *after* its potential elements.

In this **cumulative hierarchy style construction**, **no set ever contains itself**. That's exactly what ZFC enforces with the **Axiom of Foundation**.

Axiom of Foundation

- **At successor levels:** New stage is built as a powerset only from the previous level.
- **At limit levels:** New stage is built as a union from all earlier levels [we will skip but is related to infinity concept i.e., the level after infinity or a level where predecessor cannot be identified].

Ordinals up to $\omega \cdot 3 + 2$



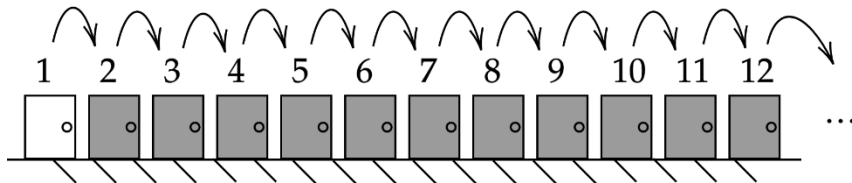
How do we compare sets?

- For two finite sets, this is easy, just count the number of elements and compare them. In general, we can compare sets by:
- **Equality** ($A=B$).
- **Inclusion** ($A\subseteq B$).
- **Size (cardinality)** ($|A|=|B|$).
- **Membership** ($A\in B$).
- **Rank (in ZFC)** (when they appear in the hierarchy).
- $\subseteq$ compares all elements of one set to another, $\in$ asks whether the set itself (as one object) belongs inside another set.
- Take $A=\{2\}$, $B=\{1,2,3\}$:
- $A\subseteq B$? Yes , because 2 (the only element of A) is in B.
- $A\in B$? No , because B's elements are numbers (1,2,3), not the set $\{2\}$.

How do we compare sets?

- What about two infinite sets?
- Example: {set of all even natural numbers} vs {set of all odd natural numbers}
- Turns out we need functions...

Hilbert's hotel



- Suppose there is a hotel with infinitely many rooms.
- And suppose they are all full.
- Can you accommodate 1 or finitely many more guests, by shifting around the existing guests?
- What if infinitely many more guests arrive?

Functions

Let A, B be two sets. A **function** f from A to B is an assignment of exactly one element of B to each element of A .
i.e., $f : A \rightarrow B$ is a subset R of $A \times B$ such that

- i. $\forall a \in A, \exists b \in B$ such that $(a, b) \in R$ [Every a has **at least one** b] and
- ii. if $(a, b) \in R$ and $(a, c) \in R$, then $b = c$ [and **at most one** b].

- We write $f(a) = b$ and call b the **image** of a .
- $\text{Range}(f) = \{b \in B \mid \exists a \in A \text{ s.t. } f(a) = b\}$, $\text{Domain}(f) = A$

□ Composition of functions

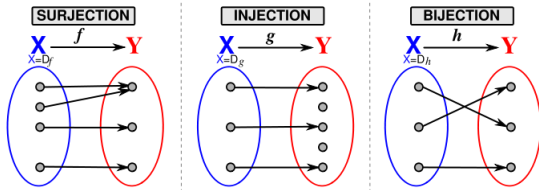
- If $g : A \rightarrow B$ and $f : B \rightarrow C$, then $f \circ g : A \rightarrow C$ is defined by $(f \circ g)(x) = f(g(x))$.
- Defined only if $\text{Range}(g) \subseteq \text{Domain}(f)$.

Functions

- Let's take: $A=\{1,2,3\}$, $B=\{a,b,c\}$, $C=\{X,Y\}$
- Define: $f=\{(1,a),(2,b),(3,c)\}$, $g=\{(a,X),(b,X),(c,Y)\}$
- Then: $f(1)=a$, $f(2)=b$, $f(3)=c$.
- $g(a)=X$, $g(b)=X$, $g(c)=Y$.
- So composition: $(g \circ f)=\{(1,X),(2,X),(3,Y)\}$.

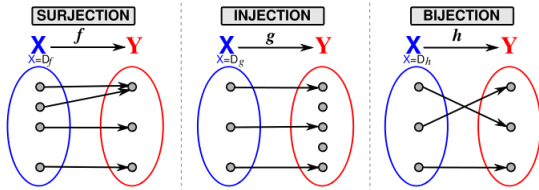
- If $f(x) = x^2$, $g(x) = x - x^3$ with $f, g: \mathbb{R} \rightarrow \mathbb{R}$, what is $f \circ g(x)$, $g \circ f(x)$?
- Composition of functions is associative
 - If $h: A \rightarrow B$ and $g: B \rightarrow C$ and $f: C \rightarrow D$, then
 - $f \circ (g \circ h) = (f \circ g) \circ h$.
- Inverse of a function
 - If $f: A \rightarrow B$ is a function i.e., $f(a) = b$, then $f^{-1}: B \rightarrow A$ defined by $f^{-1}(b) = a$, is called its inverse.

Properties of Functions



- **Surjective or onto:** $f: X \rightarrow Y$ is surjective if $\forall y \in Y, \exists x \in X$ such that $f(x) = y$.
- **Injective or 1-1:** $f: X \rightarrow Y$ is injective if $\forall x_1, x_2 \in X$, if $f(x_1) = f(x_2)$, then $x_1 = x_2$.
- **Bijective or 1-1 correspondence:** A function is bijective if it is surjective and injective.
- If f is a bijection, then its inverse function exists and $f \circ f^{-1} = f^{-1} \circ f = \text{id}$ [identity function i.e., returns the input]

Finite vs. infinite sets



- **Surjective or onto:** $f: X \rightarrow Y$ is surjective if $\forall y \in Y, \exists x \in X$ such that $f(x) = y$.
 - If X, Y finite, $|X| \geq |Y|$
- **Injective or 1-1:** $f: X \rightarrow Y$ is injective if $\forall x_1, x_2 \in X$, if $f(x_1) = f(x_2)$, then $x_1 = x_2$.
 - If X, Y finite, $|X| \leq |Y|$
- **Bijective or 1-1 correspondence:** A function is bijective if it is surjective and injective.
 - If X, Y finite, $|X| = |Y|$

Finite vs. infinite sets

F1: $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) = x^2$

F2: $f : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$ such that $f(x) = x^2$.

F1 is not Surjective: Only 1,4,9... have a mapping from x.
Rest don't.

F1 is not Injective: $f(2) = f(-2) = 4$ but $2 \neq -2$.

F1 is not Bijective: Neither Surjective nor Injective

F2 is Surjective: Every number in y has a corresponding $\sqrt{}$
(root 2) value in x.

F2 is Injective: No such x_1 and x_2 where $f(x_1) = f(x_2)$ except
when $x_1 = x_2$.

F2 is Bijective: Since both Surjective and Injective

Properties of finite and infinite sets

Relative notion of “size”

- Two finite or infinite sets have the same “size” iff there is a bijection between them.

Similarities between finite and infinite sets

- Let **bij** be bijection from i to j
- $\exists$ **bij** from A to B and B to C , implies $\exists$ **bij** from A to C .
- $\exists$ **bij** from A to B , then $\exists$ **bij** from B to A .
- **Schröder-Bernstein Theorem:** $\exists$ **surj** from A to B and $\exists$ **surj** B to A , implies $\exists$ **bij** from A to B .

Differences between finite and infinite sets

- For finite sets, if A is a set and $b \notin A$, then $|A| \neq |A \cup \{b\}|$.
- What about infinite sets ?

Difference between finite and infinite sets

Theorem

Let A be a set and $b \notin A$. Then A is infinite iff there is a bijection from A to $A \cup \{b\}$.

Lets starts with b to a ($\Leftarrow$)

1. Show contrapositive! If A is finite, then there can't be a bijection from A to $A \cup \{b\}$.

Difference between finite and infinite sets

Theorem

Let A be a set and $b \notin A$. Then A is infinite iff there is a bijection from A to $A \cup \{b\}$.

Let's see the more difficult direction ($\Rightarrow$)

1. If A infinite, then $A \neq \emptyset$, so let $a_0 \in A$. Define $f(a_0) = b$.
2. Now $A \setminus \{a_0\}$ is infinite $\Rightarrow$ non-empty, so let $a_1 \in A \setminus \{a_0\}$. Define $f(a_1) = a_0$.
3. $\forall i \in \mathbb{N}, i \geq 1$, $A \setminus \{a_0, \dots, a_{i-1}\}$ is infinite, hence non-empty, so let $a_i \in A \setminus \{a_0, \dots, a_{i-1}\}$. Define $f(a_i) = a_{i-1}$.
4. Collecting all such a_i 's, we get a subset $A' = \{a_i \in A \mid i \in \mathbb{N}\} \subseteq A$. (Note it may be that $A \neq A'$).
5. Now, $\forall a \in A$ if $a \notin A'$, we define $f(a) = a$.
6. f is a bijection!

Difference between finite and infinite sets

Corollary: Difference between finite vs infinite sets

- If A, B are infinite and $A \subset B$, there can still be a bijection from A to B , i.e., they have the same “cardinality”.
- From any set A which is infinite, there is a surjection from A to $\mathbb{N}$.

Comparing infinite sets using functions

Theorem

There is a bijection from $\mathbb{Z}$ to $\mathbb{N}$.

Proof: Hilbert's hotel argument. But how do you formalize it?

$$f(x) = \begin{array}{ll} -2x & \text{if } x \leq 0 \\ 2x - 1 & \text{else} \end{array}$$

Some questions...

1. Is there a bijection between $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$?
2. Is there a bijection between $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$?

Countable and countably infinite sets

Definition

- For a given set C , if there is a bijection from C to $\mathbb{N}$, then C is called **countably infinite**.
- A set is **countable** if it is finite or countably infinite.
Examples: even numbers, number of horses,...
- Countably infinite sets are the “smallest” infinite sets.
- What are the other properties of countable sets?

Unions of countable sets is countable

Let $A = \{a_0, \dots\}$ be a countably infinite set and B be a set. Is $A \cup B$ countable, under the following conditions?

1. $B = \{b_0\}$ is a singleton
2. $B = \{b_0, \dots, b_n\}$ is a finite set
3. $B = \{b_0, \dots\}$ is a countably infinite set
Can we say $\{a_0, \dots, b_0, \dots\}$ is a countably infinite set?

- But then what is the position of b_i (i.e., natural number corresponding to it)?
- Rather, choose $\{a_0, b_0, a_1, b_1, \dots\}$, then b_i is at $(2i + 1)^{th}$ position.
- Formally, define a bijection $f : (A \cup B) \rightarrow \mathbb{N}$ by $f(a_i) = 2i$ and $f(b_i) = 2i + 1$
- Overall, union of countably infinite set stays countably infinite

Products of countable sets are countable

Theorem: Cartesian product of two countably infinite sets is countably infinite

Proof: Let A, B be countably infinite. Find a way to “number” the elements in $A \times B = \{(a, b) \mid a \in A, b \in B\}$.

- That is, define a bijection from $A \times B$ to $\mathbb{N}$.
- $f(a_i, b_j) = \left(\sum_{k=1}^{i+j} k\right) + j + 1$ [Cantors diagonal enumeration]

$i \setminus j$	0	1	2	3	4
0	1	3	6	10	15
1	2	5	9	14	
2	4	8	13	19	
3	7	12	18		
4	11	17			
5	16				

Countable sets and functions

Are the following sets countable?

- Set of all integers $\mathbb{Z}$
- $\mathbb{N} \times \mathbb{N}$
- Set of rational numbers $\mathbb{Q}$
- Set of all real numbers $\mathbb{R}$ [No]